Twisted Pair Cable and Co-axial Cable

# GUIDED OR BOUNDED OR WIRED TRANSMISSION MEDIA

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# WHAT IS GUIDED TRANSMISSION MEDIA?

- A medium where signals are transmitted through a physical pathway like cables or wires
- Provides a controlled and predictable environment for data transmission
- Examples: Twisted Pari Cable, Co-axial Cable, Optical Fiber
- Highly reliable for short-to-medium distances
- Used in both analog and digital communication

# TWISTED PAIR CABLE

- A type of cable with two insulate copper wires twisted together to reduce electromagnetic interference and crosstalk
- One wire of the pair is used for receiving data signal and the other wire is used for transmitting data
- Common in telecommunication and networking

# CATEGORIES OF TWISTED PAIR CABLES

**Twisted Pair Cables are classified by EIA (Electronics Industries Alliance) into seven categories:**

- **Category 1: UTP used in telephone lines with data rate < 0.1 Mbps**
- **Category 2: UTP used in transmission lines with data rate of 2 Mbps**
- **Category 3: UTP used in LANs with a data rate of 10 Mbps**
- **Category 4: UTP used in Token Ring networks with a data rate of 20 Mbps**
- **Category 5: UTP used in LANs with a data rate of 100 Mbps**
- **Category 6: UTP used in LANs with a data rate of 200 Mbps**
- **Category 7: UTP used in LANs with a data rate of 10 Mbps**

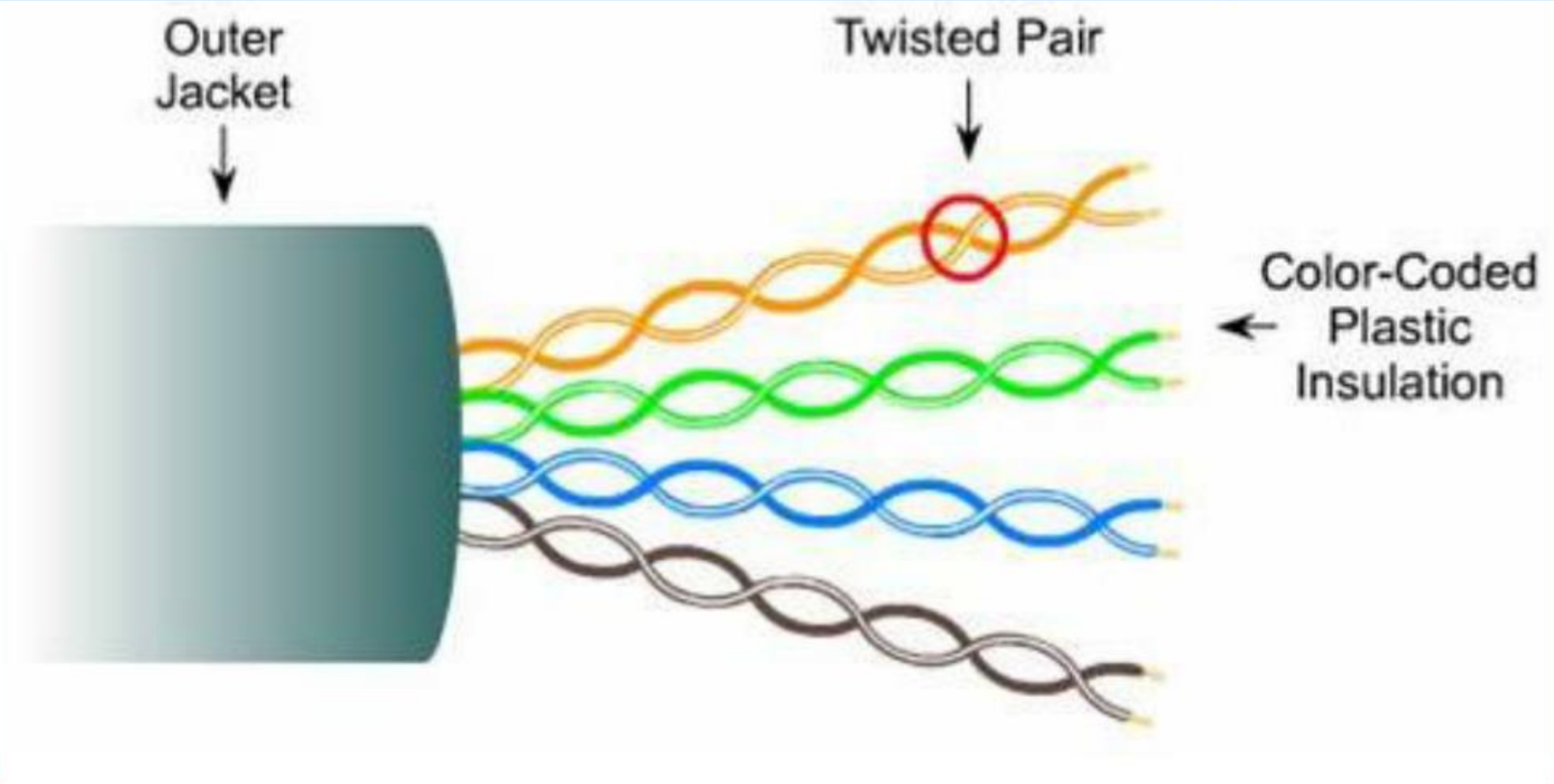
# TYPES OF TWISTED PAIR CABLE

- It is of two types:

  - Unshielded Twisted Pair (UTP) Cable
  - Shielded Twisted Pair (STP) Cable

# UTP

- No additional shielding; just twisted pairs covered by an outer insulating jacket
- Has lower bandwidth and it may be interfered by external sources
- Mainly used in telephone connections



# ADVANTAGES OF UTP

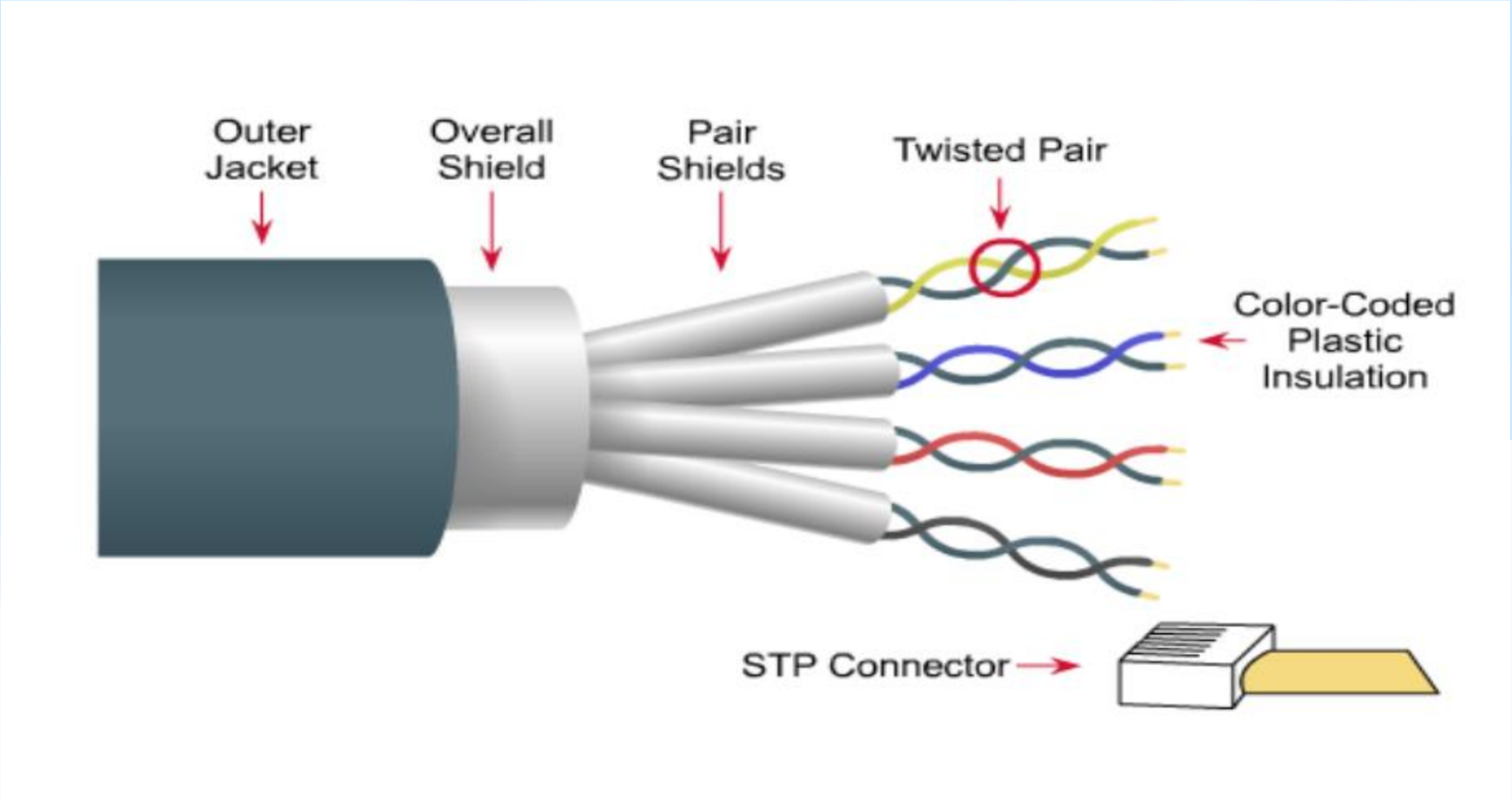
- Cost-effective
- Suitable for transmitting both data and voice via UTP Cable
- Designed to reduce crosstalk, RFI (Radio-Frequency Interference), and EMI
- Considered as faster copper-based data transmission cable
- Installation of UTP is easier

# DISADVANTAGES OF UTP

- Can only be used in length segment up to 100 meters
- Has limited bandwidth for transmitting the data
- Does not provide a secure connection for data transmitting over a network
- More susceptible to EMIs, RFIs, and cross-talks in compared to STP

# STP

- Twisted pairs with an additional shielding layer (e.g., metallic foil or braid)
- Electromagnetic noise penetration is prevented by metal casing and this shielding also eliminates crosstalk
- Widely used in LAN for digital data transmission



# ADVANTAGES OF STP

- Lower noise and attenuation than UTP
- It is shielded with a plastic cover that protects the STP cable from a harsh environment and increases the data transmission rate
- Reduces the chances of crosstalk and protects from external interference

# DISADVANTAGES OF STP

- More expensive than UTP
- Requires more maintenance to reduce the loss of data signals
- No segment improvement in length despite its thick and heavier connection
- Usually used as a grounded wire

# STRUCTURE OF TWISTED PAIR CABLES

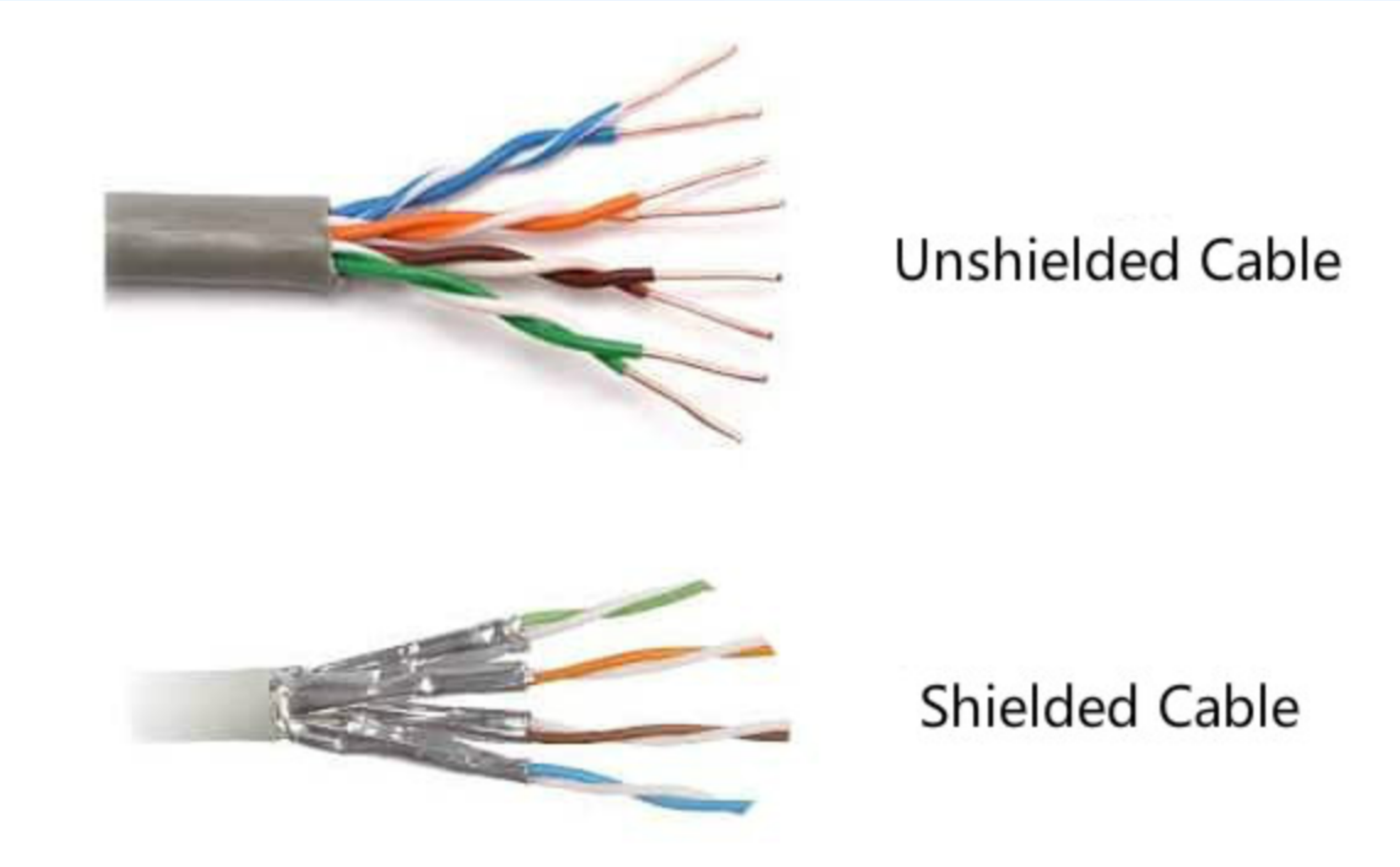
- **Copper Conductors**
  - Two insulated copper wires twisted together
  - Purpose: Transmit electrical signals

- **Insulation**
  - Each wire is coated with a plastic insulating material
  - Purpose: Prevents signal leakage and cross-talk between wires

- **Twisting of Wires**
  - Wires are twisted to cancel out EMI and cross-talk
  - The no. of twists per unit length affects the cable's performance (more twists = less interference)

- **Optional Shielding (in STP)**
  - Metallic shield (foil or braided) around the twisted pairs
  - Purpose: Provides additional protection against EMI and RFI

- **Outer PVC Jacket**
  - Protects the cable from physical damage and environmental factors



# APPLICATIONS OF TWISTED PAIR CABLE

- **Telecommunication**
  **- Commonly used in telephone lines**
- **Local Area Networks (LANs)**
  **- Used to connect computers and other devices**
- **DSL Connections**
  **- Delivers high-speed internet over telephone lines**
- **Voice and Data Transmission**
  **- Used in PBX (Private Branch Exchange) systems and Ethernet networks**

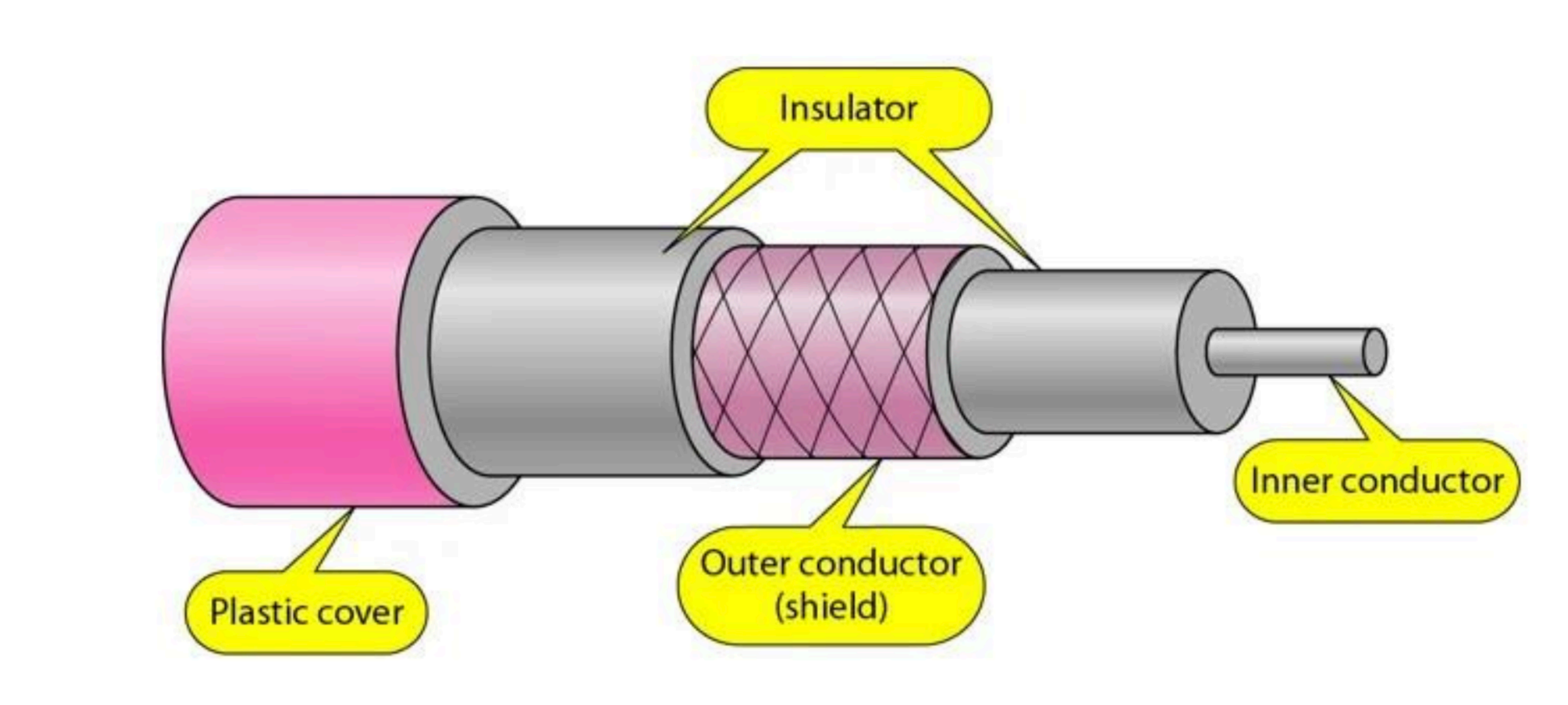
# CO-AXIAL CABLE

- Type of cable with a central conductor surrounded by insulating material, a metallic shield, and an outer jacket
- Designed for high-frequency signal transmission
- Offers excellent resistance to EMI
- Widely used in cable TV networks, internet connections, and telecommunications
- Transmits information in two modes
  - Baseband mode (dedicated cable bandwidth)
  - Broadband mode (cable bandwidth is split into separate ranges)

# STRUCTURE

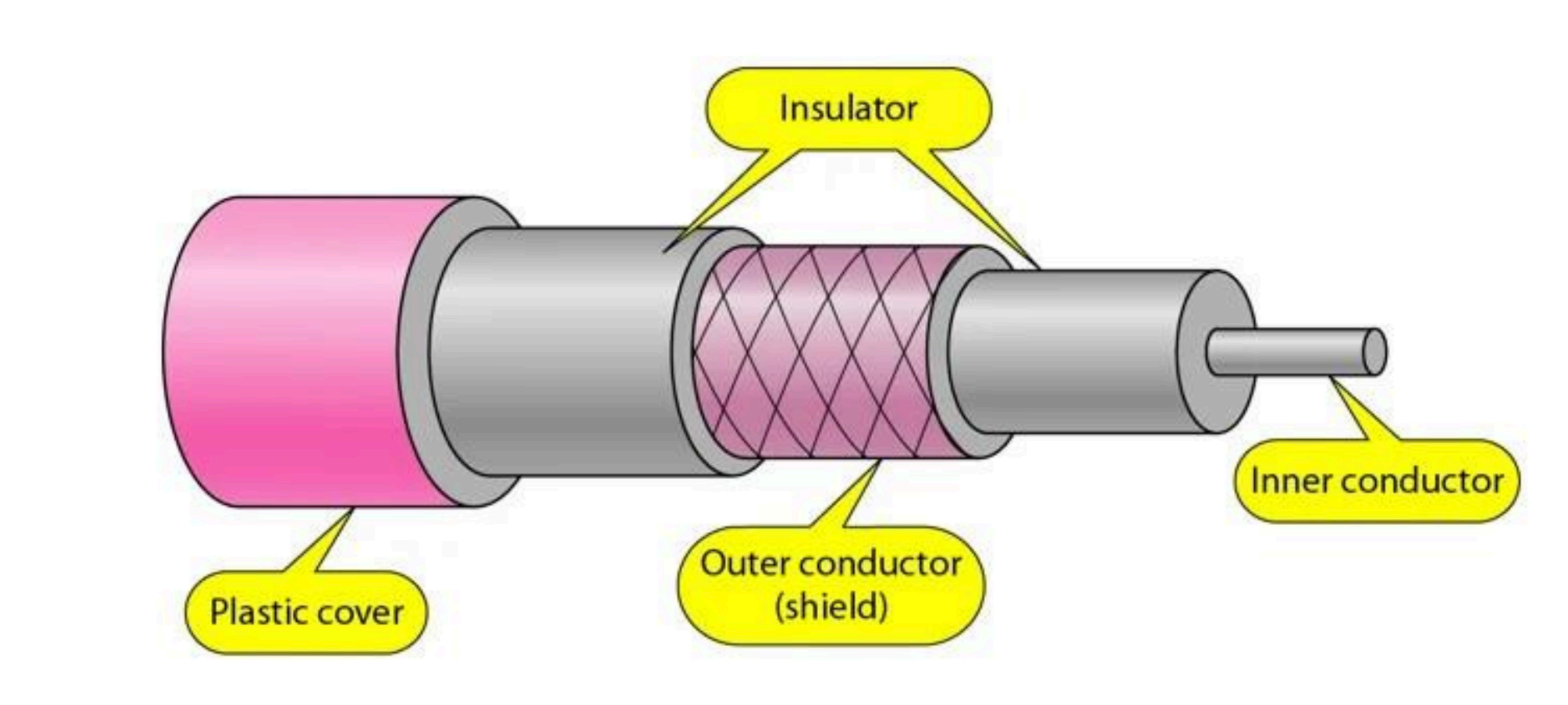
- **Inner Conductor**
  **- A single solid wire or stranded conductor made of copper or aluminum**
  **- Purpose: Transmits the electrical signal**

- **Dielectric Insulator**
  **- A non-conductive insulating material (e.g., polyethylene or foam)**
  **- Purpose: Provides spacing and insulation between the inner conductor and the outer shield**



# STRUCTURE

- **Outer Shield (Metallic Shield)**
  - Made of braided copper or aluminum foil
  - Purpose: Protects against EMI and RFI by acting as a shield




- **Outer PVC Jacket**
  - The outermost layer made of PVC or similar materials
  - Purpose: Protects the cable from physical damage, moisture, and chemical exposure

# ADVANTAGES OF CO-AXIAL CABLE

- **High Bandwidth**
  - Suitable for high-speed internet and HD video signals
- **Shielding**
  - Reduces EMI and RFI, ensuring a stable signal
- **Durability**
  - Resistant to physical wear and environmental factors
- **Long Distance**
  - Can transmit signals over long distances with minimal loss (compared to twisted pair cable)

# DISADVANTAGES OF CO-AXIAL CABLE

- **Cost**
  **- More expensive than twisted pair cables due to complex design**
- **Installation**
  **- Difficult to install and maintain due to its thickness and rigidity**
- **Signal Loss**
  **- Experiences attenuation over long distances, requiring signal boosters or amplifiers**